Gennadii Filatov

Saint Petersburg, Russia • Erasmus+ exchange at UAIC, Iaşi, Romania

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Summary

M.Sc. student in machine learning working on the evaluation of synthetic tabular data and reliable model ranking (AISTATS 2026 workshop paper). Currently strengthening my background in Bayesian statistics and probabilistic programming. Seeking a PhD position starting in 2027 on model comparison and reliable inference under misspecification.

Research interests: calibration and model comparison; synthetic tabular data; uncertainty quantification; probabilistic machine learning; trustworthy generative modeling

Education

ITMO University, St. Petersburg, Russia Sep 2025 – Jun 2027 (expected)

M.Sc. in Applied Mathematics and Computer Science, Big Data and Machine Learning (taught in English) GPA: 4.00/4.00 (maximum)

- Thesis: calibration of synthetic data for trustworthy model selection; supervisors: Prof. Dr. Alexey Zaytsev; Dr. Irina Deeva.
- *Key courses (selected):* Multivariate Data Analysis; Neural Network Architectures; Optimization Methods; Advanced Machine Learning Technologies *(in progress)*.

Alexandru Ioan Cuza University of Iaşi (UAIC), Romania Sep 2026 – Present

***Erasmus+** Academic Exchange, Faculty of Computer Science (taught in English)* Selected participant

- *Key courses (selected):* Computational Statistics; Probabilistic Programming and Modelling; Deep Neural Networks and Generative AI *(in progress)*.

Peter the Great St. Petersburg Polytechnic University (SPbPU), Russia Sep 2021 – Jun 2025

B.Sc. in Physics with honors, Nuclear and Elementary Particle Physics (taught in Russian) GPA: 4.92/5.00 (maximum)

- Thesis: precision measurement of cathode-wire positions in the high-pressure proportional-chamber system for the PRES experiment; supervisor: Prof. Dr. Dmitriy Kotov.
- *Key courses (selected):* Probability Theory and Stochastic Processes; Statistical Methods of Experimental Data Processing; Numerical Methods.

Research Experience

NSS Lab, ITMO University Oct 2025 – Present

Machine Learning Researcher (supervisor: Dr. Irina Deeva) *Code*Saint Petersburg, Russia

- Developed two calibration approaches for synthetic model evaluation: a constrained quadratic program that aligns synthetic and real loss vectors, and a Bayesian Personalized Ranking (BPR)-style pairwise objective that directly calibrates model orderings.
- Evaluated the methods on 10 tabular datasets, 122 model configurations, and 5 generative paradigms; improved mean Spearman correlation with real-data rankings from 0.37 to 0.73 for classification and from 0.53 to 0.80 for regression, resulting in an AISTATS 2026 workshop publication.
- Currently extending the framework to unseen model families and limited real-evaluation budgets, with selection-regret metrics, calibratability diagnostics, and shrinkage or abstention mechanisms for preventing harmful calibration.

SMILES-2026 International Summer School on Machine Learning Jul 2026

Selected Research Participant, Team Project (104 selected from 2,500+ applicants, ∼4% acceptance rate) Suzhou, China

- Contributed to *Block Categorical Flow Maps*, a few-step text-generation method; led a grid study over function evaluations and block size, and the team reduced generation perplexity from 166 to 49 in 16 sampling steps.

High Energy Physics Department, SPbPU Jul 2023 – Aug 2025

Senior Laboratory Assistant (supervisor: Prof. Dr. Dmitriy Kotov) Saint Petersburg, Russia

- Developed image-processing methods for multi-wire proportional chambers in the PRES experiment, analysed systematic measurement errors, and improved the precision of cathode-wire position reconstruction by 30%.

Laboratory of Diagnostics of High-Temperature Plasma, SPbPU Feb – Jun 2023

Machine Learning Research Intern (supervisor: Prof. Dr. Alexander Yashin) Saint Petersburg, Russia

- Developed a 1D CNN (TensorFlow/Keras) for automated detection of filaments in tokamak diagnostic signals.

Selected Publications

Workshop paper: **G. Filatov**, I. Deeva. *"When Synthetic Data Is Enough: Calibration for Tabular Model Ranking."* Workshop on Towards Trustworthy Predictions: Theory and Applications of Calibration for Modern AI, **AISTATS 2026**. [OpenReview] [Code]

Selected Presentations

- **Talk:** *Calibration of synthetic tabular data for trustworthy model selection.* Faculty seminar, Artificial Intelligence Technologies Faculty, ITMO University, May 2026.
- **Poster:** *When Synthetic Data Is Enough: Calibration for Tabular Model Ranking.* AISTATS 2026 Workshop, Morocco.
- **Poster:** *Calibration of synthetic tabular data for correct model selection.* XV Congress of Young Scientists, ITMO University, 2026.

Honors & Awards

- **National Academic Scholarship for Science and Education, 2026–2027** — competitive merit-based scholarship awarded by ITMO University.
- **Erasmus+ Mobility Grant, 2026–2027** — competitive EU-funded exchange at Alexandru Ioan Cuza University of Iaşi, Romania.
- **Enhanced State Academic Scholarship, 2022–2025** — awarded to students in the top 5% at SPbPU.
- **"Student of the Year" SPbPU, 2024** — university-wide distinction for academic, research, and extracurricular achievement.

Technical Skills

Programming: Python, C++, Bash, LaTeX, MATLAB
ML / statistics: PyTorch, scikit-learn, TensorFlow/Keras, NumPy, SciPy, pandas, Optuna
Research tools: Git, Docker
Languages: Russian (native), English (CEFR B2; ITMO English Language Level Examination, 2025)

Leadership, Teaching & Academic Service

AI Challenge Sep – Nov 2026
Solution Evaluator
Evaluated main- and final-stage solutions in an international AI competition (21.5k registered participants from 114 countries); reviewed code reproducibility and originality and took part in live code-review sessions.

Summer Scientific School "A5 School" Aug 2022 – Jul 2024
Co-Founder & Head of Education
Founded a science summer school, trained tutors, and designed the curriculum; 115+ school students completed the program.

International Association of Physics Students (IAPS) Local Committee Saint Petersburg Apr 2023 – Apr 2025
President
Led a student outreach team; secured three grants totaling EUR 17,000 and organized 113+ events reaching 1,600+ students.

Referees

Dr. Irina Deeva
M.Sc. thesis, NSS Lab
ITMO University, HSE University
ideeva@hse.ru

Prof. Dr. Alexey Zaytsev
Research supervisor
Applied AI lab, BIMSA
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Prof. Dr. Dmitriy Kotov
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Interests

Regular cycling (∼60 km per week); hiking; landscape photography.